DS	**Session**	Searching	**Question Information**	Level 1 Challenge 10

Question Description:
An array A contains integers with the following constraints:

'A' contains elements in sorted order.
Integer i occurs i*floor(sqrt(i))+ceil(i/2) times in the array.
All elements are greater than or equal to 1.

Fathima has given Q queries of type:

L R: Find the number of distinct values present in subarray A[L...R].
Note: 1-based indexing is followed.

Constraints:
1 <= Q <= 10^5
1 <= L <= R <= 10^13

Input Format:
The first line contains an integer Q denoting the number of queries.
Next Q lines contains two space-separated integers L, R, denoting the query.

Output Format:
Print the output in separate lines contains required number of distinct values.

Example

Sample Input:

2

1 3

1 6

Sample Output:

2

3

Explanation

First few elements of the array A are $1, 1, 2, 2, 2, 3, 3, 3, 3, 3, \ldots$

- For Query 1:-
  - Number of distinct elements in subarray $A[1...3]$ is 2.
- For Query 2:-
  - Number of distinct elements in subarray $A[1...6]$ is 3.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
3 3
2 6
```

EXPECTED OUTPUT

```
1
3
```

Test Case 2

INPUT (STDIN)

```
3
3 3
2 6
5 4
```

EXPECTED OUTPUT

```
1
3
1
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
while(l<ans1)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
6
```

Test Case 2

TOKEN COUNT

```
400
```

Test Case 3

NLOC

```
56
```